

Enrico FOGLIA,

PhD candidate @ ISAE-SUPAERO and U. de Sherbrooke

✉ enricofoglia@proton.me

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Professional summary

Computational scientist & machine learning engineer specializing in generative modelling, deep learning and uncertainty quantification. Experience with high-fidelity aerodynamics simulations (~ 100 millions cells) on large supercomputers on multiple GPUs.

Skills

Languages	🍃 Italian (native), English (fluent), French (fluent), Spanish (fluent), Portuguese (basic).
Machine Learning	🍃 PyTorch , uncertainty quantification (UQ), TensorFlow, scikit-learn. Graph and convolutional neural networks.
HPC	🍃 GPU-acceleration (OpenAcc), git, GitHub and GitLab, bash. Experience working on large HPC clusters (Trillium, Setonix).
Scientific Computing	🍃 Python , FORTRAN (modern and 77), C++, scipy, numpy, matplotlib, Paraview. Adjoint methods.
Domain Expertise	🍃 Fluid dynamics , machine learning , aeroacoustics, probability theory, stochastic systems.
Misc.	🍃 Academic research, teaching, training, $\text{\LaTeX}$ typesetting and publishing.

Research Experience

2023 - ongoing	🍃 PhD candidate. DAEP, ISAE-SUPAERO and Mechanical Engineering Department, Université de Sherbrooke. • Developed probabilistic generative models for turbulent fluctuations. • Developed new UQ techniques for deep learning. • Executed DNS simulations involving $\sim 10^8$ cells on international HPC clusters (Trillium, Setonix).
2023	🍃 Intern. AKOU, DAAA, ONERA: Engineered a Python package for adjoint-based aeroacoustic optimization.
2021-2023	🍃 Graduate researcher. DAEP, ISAE-SUPAERO: Built Reduced Order Models using polynomial and time-delay coordinates for airfoil pitching
summer 2022	🍃 Intern. DAEP, ISAE-SUPAERO: Integrated FORTRAN code with python interface for airfoil aerodynamics simulation tool.

Education

2021 – 2023	🍃 M.Sc., Aerospace Engineering ISAE-SUPAERO Major in Advanced Aerodynamics and Propulsion. Average GPA: 4.0
2018-2021	🍃 B.Sc. Aerospace Engineering Politecnico di Torino Part of the "Young Talents" program (Honors Course). Final grade 110/110.
2021	🍃 Erasmus Universidad Politécnica de Madrid

Research Publications

Journal Articles

- 1 E. Foglia, B. Bobbia, M. Bauerheim, T. Jardin, S. Moreau, and P. Fua, "Empirical uncertainty quantification in deep learning: Applications to fluid dynamics," *Computers & Fluids*, p. 107 117, 2026.

Conference Proceedings

- 1 E. Foglia, J. Rendon, S. Moreau, M. Bauerheim, and T. Jardin, "Analysis of airfoil self-noise sources using wavelet-based denoising," in *32nd AIAA/CEAS Aeroacoustics Conference (2026)*, 2026, p. 3237.
- 2 E. Foglia, M. Daroukh, M. Buszyk, I. Salah El-Din, and S. Moreau, "Towards the optimization of propeller tonal noise in the frequency domain using the discrete adjoint method," in *30th AIAA/CEAS Aeroacoustics Conference (2024)*, 2024, p. 3153.

Reference

For referrals, please contact Prof. Michaël Bauerheim at Michael.BAUERHEIM@isae-supero.fr